Technical Refinements for Exact Hoeffding Constants

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Abstract

This supplement contains the full proofs of the fixed-sample success-count extremality and exact imbalance response, analytic contraction on compact success-proportion intervals, and auxiliary pointwise-curvature examples. The main paper proves the parity-dependent exact Hoeffding constants independently of these refinements. The symbols and equation references used here are those of the accompanying manuscript.

1 Uniform extremizers at fixed sample size

At each fixed sample size, the following results maximize the proxy over every success count and compute its local separation from the Bernoulli envelope.

1.1 A two-parameter analytic family

Fix an integer $m \geq 1$. Write $e = N^{-1}$, $d = 1 - 2k/N$, $p = (1-d)/2$, and $q = 1-p$. For $i = 0, \ldots, m$, define

$$P_i(e,d) = \binom{m}{i} \frac{\prod_{r=0}^{i-1}(p - re)\prod_{r=0}^{m-i-1}(q - re)}{\prod_{r=0}^{m-1}(1 - re)}. \tag{1}$$

Empty products equal one. For physical parameters this is the hypergeometric probability mass function. Vandermonde's falling-factorial identity gives $\sum_i P_i = 1$ and $\sum_i iP_i = mp$. In a rectangle around $(e,d) = (0,0)$, every factor is positive and the masses are analytic. At $e = 0$ the law is Binomial(m,p). Define

$$\psi(e,d,t) = \log \sum_{i=0}^{m} P_i(e,d) e^{t(i-mp)}, \qquad Q(e,d,t) = \frac{2\psi(e,d,t)}{t^2},$$

with $Q(e,d,0) = \partial_t^2 \psi(e,d,0)$. Division by t^2 is removable because $\psi(e,d,0) = \partial_t \psi(e,d,0) = 0$. Let

$$\mathcal{R}_m(e,d) = \frac{4}{m(1-me)} \max_{t \in \mathbb{R}} Q(e,d,t).$$

Lemma 1.1 (Joint analytic optimizer)**.** *For each fixed m, there is a rectangle around $(0,0)$ on which $Q(e,d,\cdot)$ has a unique global maximizer $\tau_m(e,d)$. Both τ_m and $\mathcal{R}_m$ are real analytic. Moreover,*

$$\tau_m(e,-d) = -\tau_m(e,d), \qquad \mathcal{R}_m(e,-d) = \mathcal{R}_m(e,d), \tag{2}$$

$$\tau_m(e,0) = 0, \qquad \mathcal{R}_m(e,0) = \frac{1}{1-e}. \tag{3}$$

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Proof. At $(e,d) = (0,0)$, $Q(t) = 2m \log\cosh(t/2)/t^2$ has its unique maximum at zero and $Q_{tt}(0) = -m/48$. For nearby parameters, $|H - mp| \le m$ gives $0 \le Q(t) \le 2m/|t|$. Choose T so large that this tail bound lies strictly below $m/4$. On $[-T,T]$, joint continuity, including at $t = 0$, yields uniform convergence to the limiting objective. Its strict gap outside any neighborhood of zero forces every maximizing point into that neighborhood. Joint analyticity and $Q_{tt}(0,0,0) < 0$ then give a fixed smaller neighborhood where $Q_{tt} < 0$. The analytic implicit-function theorem applied to $Q_t = 0$ gives one critical point there, which the localization argument identifies as the unique global maximum.

The identity $P_i(e,-d) = P_{m-i}(e,d)$ implies $Q(e,-d,t) = Q(e,d,-t)$, proving (2). At $d = 0$ the objective is even in t, so uniqueness gives $\tau_m(e,0) = 0$. Its value is the exact variance

$$\partial_t^2 \psi(e,0,0) = \frac{m}{4}\frac{1-me}{1-e},$$

which proves (3). □

Theorem 1.2 (Uniform fixed-sample maximum)**.** *For each fixed integer $m \ge 1$, there exists $N_0(m)$ such that, for every odd $N \ge N_0(m)$,*

$$\max_{1\le k<N} \frac{4N}{m(N-m)} \sigma^2_{\mathrm{opt}}(H_{N,k,m}) = R_{N,1,m}. \tag{4}$$

The maximizing success counts are exactly $k = (N-1)/2$ and $k = (N+1)/2$. If $m \ge 2$, this maximum is strictly below L_N. Consequently, for every fixed $M \ge 2$, all sample sizes $2 \le m \le M$ and all success counts are simultaneously below L_N once N is sufficiently large.

Proof. At $e = 0$, the Bernoulli proxy formula gives

$$\mathcal{R}_m(0,d) = g(d) := \frac{d}{\operatorname{artanh} d}, \qquad g(0) = 1.$$

In particular $g(d) = 1 - d^2/3 + O(d^4)$, so

$$\partial_d^2 \mathcal{R}_m(0,0) = -\frac{2}{3}.$$

Joint analyticity makes this second derivative continuous in (e,d). Choose $e_0, d_0 > 0$ inside the analytic rectangle such that

$$\partial_d^2 \mathcal{R}_m(e,d) < 0 \qquad (0 \le e \le e_0,\ |d| \le d_0).$$

Evenness in d gives $\partial_d \mathcal{R}_m(e,0) = 0$. The mean value theorem now yields $\partial_d \mathcal{R}_m(e,d) < 0$ for $0 < d \le d_0$, uniformly for $0 \le e \le e_0$. Thus $\mathcal{R}_m(e,\cdot)$ is strictly decreasing on $[0,d_0]$, and among odd-population success counts in this interval the maximum is uniquely attained at $d = 1/N = e$.

For the remaining success counts, the elementary-symmetric-mean inequality gives

$$\mathbb{E}e^{tH_{N,k,m}} \le (q + pe^t)^m. \tag{5}$$

Indeed, $\mathbb{E}e^{tH}$ is the normalized mth elementary symmetric polynomial of N positive weights, k equal to e^t and the others equal to one. At fixed sum, averaging a pair of weights increases that polynomial by $(a-b)^2 e_{m-2}(\text{rest})/4 \ge 0$. Repeatedly averaging the largest and smallest weights decreases the sum of their squares by $(a-b)^2/2$; consequently their difference tends to zero, and the weights converge to their common mean. The elementary symmetric polynomial never decreases during this procedure. The case $m = 1$ is equality. This proves (5). After centering,

the Bernoulli proxy formula yields

$$\frac{4N}{m(N-m)}\sigma^2_{\mathrm{opt}}(H_{N,k,m}) \le \frac{g(d)}{1-me} \le \frac{g(d_0)}{1-me} \quad (d \ge d_0).$$

Since $g(d_0) < 1$, this last bound is below one for sufficiently small $e > 0$, whereas $\mathcal{R}_m(e, e) = 1 + e + O_m(e^2) > 1$. Reflection covers negative d. This proves (4) and the equality cases. The strict comparison with L_N follows from Corollary A.2. Taking the largest of finitely many thresholds proves the assertion for $2 \le m \le M$. □

1.2 Exact local separation at finite population size

The polynomial proof of Theorem 2.3 also applies to the normal form with integer $j \ge 1$ and real $h, r \ge 0$. In this extension, the coefficients in (22) use Gamma functions for factorial ratios. Their ratio, the differential equation, and the stationary identity remain unchanged; Jacobi orthogonality holds for real $h, r \ge 0$, and the positivity certificate uses only $S = h + r \ge 0$. The cases $j \ge 3$ and the two boundary layers therefore prove the same curvature bound. This observation allows a continuous success-count perturbation at $k = N/2$, including odd N.

Theorem 1.3 (Exact imbalance coefficient)**.** *Fix integers $N \ge 5$ and $1 \le m \le \lfloor N/2 \rfloor$. Extend the hypergeometric masses to real $k = N(1-d)/2$ by (1), with $e = 1/N$. For all sufficiently small $|d|$, the optimal normalized proxy has a unique maximizing tilt, analytic in d, and*

$$\tau_m(1/N, d) = T_{N,m}d + O_{N,m}(d^3), \tag{6}$$

$$\mathcal{R}_m(1/N, d) = \frac{N}{N-1} - A_{N,m}d^2 + O_{N,m}(d^4), \tag{7}$$

where, writing $D_{N,m} = N^2 - N - 3m(N-m) > 0$,

$$T_{N,m} = \frac{4(N-2m)(N-3)(N-1)}{(N-2)D_{N,m}}, \tag{8}$$

$$A_{N,m} = \frac{N}{3(N-1)} + \frac{2N^2(N-4)(m-1)(N-m-1)}{3(N-2)^2(N-1)D_{N,m}}. \tag{9}$$

For $m \ge 2$, $A_{N,m} > A_{N,1}$. Thus the Bernoulli envelope strictly dominates the m-draw proxy for every sufficiently small nonzero real imbalance.

Proof. At $d = 0$, use the real-margin extension with $j = m$ and $h = r = N/2 - m \ge 0$. The curvature envelope equals the variance

$$B = \frac{m(N-m)}{4(N-1)}.$$

Its twice-integrated bound is strict at every nonzero t, by the analytic-gap argument in Corollary 2.4; hence $Q(1/N, 0, t)$ has its unique global maximum at zero. The masses remain positive for $|d|$ sufficiently small. Their finite support gives uniform tail control, so the localization argument in Lemma 1.1 applies at this fixed N.

Factorial moments give, at $d = 0$,

$$\kappa_2(d) = B(1 - d^2),$$
$$\kappa_3(d) = B\frac{N - 2m}{N - 2}d + O(d^3),$$
$$\kappa_4(0) = -\frac{BD_{N,m}}{2(N - 3)(N - 1)}.$$

Since $m(N - m) \le N^2/4$, $D_{N,m} \ge N(N - 4)/4 > 0$. Consequently $Q_{tt}(1/N, 0, 0) = \kappa_4(0)/6 < 0$, and the analytic implicit-function theorem gives a unique analytic maximizing tilt. If $b = B(N - 2m)/(N - 2)$ and $c = \kappa_4(0)$, the stationary equation gives $T_{N,m} = -2b/c$, proving (8). Substitution into

$$Q(d, t) = B(1 - d^2) + \frac{bd}{3}t + \frac{c}{12}t^2 + O_{N,m}(d^4) \qquad (t = O(d))$$

gives

$$\max_t Q(d, t) = B - \left(B + \frac{b^2}{3c}\right)d^2 + O_{N,m}(d^4).$$

Reflection makes the optimized value even in d, so the stated remainder follows. Multiplying by $4N/[m(N - m)]$ yields

$$A_{N,m} = \frac{N}{N - 1}\left[1 - \frac{2(N - 2m)^2(N - 3)(N - 1)}{3(N - 2)^2 D_{N,m}}\right].$$

Subtracting $N/[3(N - 1)]$ and putting over a common denominator gives exactly the positive second term of (9). The singleton proxy is $\mathcal{R}_1(1/N, d) = Ng(d)/(N - 1)$, so its quadratic coefficient is $A_{N,1} = N/[3(N - 1)]$. The asserted local domination follows from the strict coefficient difference. □

2 Analytic contraction on compact success-proportion intervals

2.1 Fixed-sample comparison

The first finite-population correction gives a comparison away from the balanced point as well.

Theorem 2.1 (Uniform Bernoulli envelope at fixed sample size)**.** *Fix $m \ge 2$ and $0 < \eta < 1/2$. There exists $N_1(m, \eta)$ such that, for all $N \ge N_1(m, \eta)$ and $\eta \le k/N \le 1 - \eta$,*

$$\sigma^2_{\mathrm{opt}}(H_{N,k,m}) \le \frac{m(N - m)}{N - 1}\,\sigma^2_{\mathrm{opt}}(\mathrm{Bern}(k/N)). \tag{10}$$

Equality holds when $k = N/2$; the inequality is strict otherwise.

Proof. By reflection let $d \in [0, 1 - 2\eta]$. At $e = 0$ the binomial proxy has its unique maximizing tilt $t_0(d) = 4\,\mathrm{artanh}\,d$, with continuous value $t_0(0) = 0$. Its second derivative there is negative: for $d > 0$, the Bernoulli proof in Appendix D gives $G''(t_0) = c - pq > 0$ and $Q''(t_0) = -2mG''(t_0)/t_0^2 < 0$; at $d = 0$ the derivative is $-m/48$. Compactness, the uniform tail bound, and the strict uniqueness at $e = 0$ allow the localization and implicit-function arguments to be applied uniformly on this interval. Thus $\mathcal{R}_m(e, d)$ is analytic on a neighborhood of its compact parameter interval for sufficiently small $|e|$.

For p, q in this compact interval, expanding (1) gives

$$P_i(e,d) = \binom{m}{i} p^i q^{m-i} \left[1 + \frac{e}{2}\left\{m(m-1) - \frac{i(i-1)}{p} - \frac{(m-i)(m-i-1)}{q}\right\}\right] + O_{m,\eta}(e^2).$$

If $Z = q + pe^t$, binomial factorial moments therefore yield

$$\psi(e,d,t) = m \log Z - mpt + \frac{em(m-1)}{2}\left[1 - \frac{pe^{2t}+q}{Z^2}\right] + O_{m,\eta}(e^2) \tag{11}$$

on compact t intervals. Analytic divisibility by t^2 permits the same expansion for Q at zero. Evaluation at the unique limiting maximizer gives, with $a = \log(q/p) = 2\,\operatorname{artanh} d$,

$$\mathcal{R}_m(e,d) = g(d) + e\left[mg(d) - (m-1)\frac{d^2}{a^2 pq}\right] + O_{m,\eta}(e^2).$$

The continuous interpretation is used at $d = 0$. Put $D_m(e,d) = \mathcal{R}_1(e,d) - \mathcal{R}_m(e,d)$. Since $\mathcal{R}_1(e,d) = g(d)/(1-e)$, this difference vanishes at $e = 0$ and at $d = 0$ for every small e. Reflection also gives $\partial_d D_m(e,0) = 0$. Taylor's formula in d, followed by the fundamental theorem of calculus in e, therefore gives the exact identity

$$D_m(e,d) = ed^2 B_m(e,d), \qquad B_m(e,d) = \int_0^1 (1-s)\int_0^1 \partial_e \partial_d^2 D_m(re, sd)\, dr\, ds.$$

The joint continuity of the differentiated objective makes B_m continuous on a neighborhood of the compact parameter interval. The first population correction identifies its boundary values as

$$B_m(0,d) = \frac{(m-1)g(d)}{d^2}\left[\frac{d}{(1-d^2)\operatorname{artanh} d} - 1\right], \qquad B_m(0,0) = \frac{2(m-1)}{3}.$$

For $d > 0$, integration gives $\operatorname{artanh} d < d/(1-d^2)$, so $B_m(0,d) > 0$. Its compact minimum is positive. Uniform continuity then gives $B_m(e,d) > 0$ for all sufficiently small $e > 0$. This proves the normalized comparison, including strictness, and multiplying through gives (10). □

Remark 2.2 (Relation to the finite-population envelope)**.** *Theorem 2.2 now gives the finite-population proxy comparison for both parities in the canonical sub-half-population range, with equality at the balanced even margin. The present analytic argument records the first finite-population correction uniformly over compact success-proportion intervals, and Theorem 1.2 identifies the maximizing success count for each fixed sample size in sufficiently large odd populations.*

3 Pointwise curvature and its weighted integral

The exact odd-population constant defines the factor

$$\rho_N = \frac{N-1}{N} L_N = \frac{1}{N \operatorname{artanh}(1/N)} < 1.$$

The corresponding pointwise inequality takes the form

$$\psi''_{N,k,m}(t) \le \rho_N \frac{m(N-m)}{4(N-1)}. \tag{12}$$

The next proposition gives an exact tilted variance above the right side of (12).

Proposition 3.1 (Exact exceedance of the constant pointwise bound)**.** *For $N = 15$, $k = 7$, $m = 3$, and $t = \log(7/6)$,*

$$\frac{4(N-1)}{m(N-m)}\psi''_{N,k,m}(t) = \frac{307887776}{308178025} > \frac{675}{676} > \rho_{15}. \tag{13}$$

This nearest-balanced population violates (12) at the stated tilt.

Proof. With $z = 7/6$, the tilted hypergeometric weights for $j = 0, 1, 2, 3$ are proportional to

$$w_j = \binom{7}{j}\binom{8}{3-j}z^j.$$

They are rational, so both the mean and the variance are rational. Direct summation gives

$$\mathbb{E}_t H = \frac{26313}{17555}, \qquad \mathrm{Var}_t(H) = \frac{197927856}{308178025}.$$

Since $m(N-m)/[4(N-1)] = 9/14$, the first equality in (13) follows by multiplying the variance by $14/9$. The first strict inequality follows from

$$\frac{307887776}{308178025} - \frac{675}{676} = \frac{111969701}{208328344900} > 0.$$

Finally, the power series for artanh at $x = 1/15$ yields

$$\operatorname{artanh}(1/15) > \frac{1}{15} + \frac{1}{3 \cdot 15^3}, \quad \text{hence} \quad \rho_{15} < \left(1 + \frac{1}{3 \cdot 15^2}\right)^{-1} = \frac{675}{676}.$$

Combining these two inequalities proves (13). □

Proposition 3.2 (Integrated curvature identity)**.** *For every centered random variable with finite moment generating function, writing $\psi(t) = \log \mathbb{E}e^{tX}$ gives*

$$\frac{2\psi(t)}{t^2} = 2\int_0^1 (1-u)\psi''(ut)\,du, \qquad t \neq 0. \tag{14}$$

The proxy objective is therefore a weighted average of curvature along the segment from zero to the chosen tilt.

Proof. Twice integrate ψ'' using $\psi(0) = \psi'(0) = 0$ and substitute $s = ut$; the formula is valid for either sign of t. □

Theorem 11.1 bounds the weighted integral (14) through the mean-dependent envelope of Lemma 11.4. Proposition 3.1 identifies a point of this variance curve above the constant threshold in (12).